LEVEL 0 · PART I

What DevOps Actually Is

0

Goal: Understand how development, infrastructure, deployment, security, monitoring and operations form one continuous lifecycle.

RECOMMENDED TIME	TOPICS
1 week (reading + mapping)	13

PRACTICE / CAPSTONE

Map the lifecycle onto a real open-source project you know: where does each stage live, and who owns it?

The DevOps lifecycle

DevOps connects software development, infrastructure, deployment, security, monitoring

and operations into one continuous lifecycle.

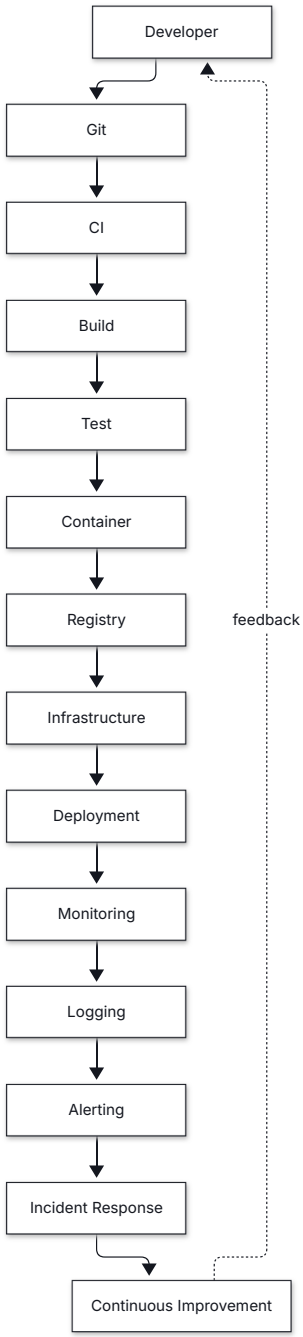


Figure 0-1

Topics

#	Topic	What you will learn
1	Software Development Lifecycle (SDLC) **p.41**	SDLC phases, Agile vs waterfall, how DevOps shortens feedback loops
2	CI/CD **p.51**	continuous integration, continuous delivery vs deployment, pipeline stages
3	Infrastructure as Code **p.63**	declarative vs imperative, version-controlled infra, Terraform/CloudFormation overview
4	Configuration Management **p.75**	Ansible/Chef/Puppet, idempotency, desired state on servers
5	Immutable Infrastructure **p.87**	replace don't patch, golden images, pets vs cattle
6	Infrastructure Automation **p.97**	automating provisioning, scripts vs tools, eliminating toil, runbooks to code
7	Deployment Strategies **p.109**	recreate, rolling, blue/green, canary, feature flags (overview; detail in Level 5)
8	High Availability and Scalability **p.121**	redundancy, SPOF, horizontal vs vertical scaling, health checks
9	Disaster Recovery **p.135**	backups, RPO/RTO, failover (overview; detail in Level 15)
10	Observability **p.147**	metrics, logs, traces, monitoring vs observability (overview; detail in Level 10)
11	Site Reliability Engineering **p.161**	SLOs, error budgets, toil, how SRE relates to DevOps (detail in Level 14)
12	DevSecOps **p.173**	shift-left security, scanning in pipelines, supply chain (detail in Level 11)
13	Platform Engineering **p.185**	internal developer platforms, golden paths, DevOps vs platform teams (detail in Level 18)
★	Level capstone project **p.199**	Map the lifecycle onto a real open-source project you know: where does each stage live, and who owns it?

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

1 Software Development Lifecycle (SDLC)

2 CI/CD

3 Infrastructure as Code

4 Configuration Management

5 Immutable Infrastructure

6 Infrastructure Automation

7 Deployment Strategies

8 High Availability and Scalability

9 Disaster Recovery

10 Observability

11 Site Reliability Engineering

12 DevSecOps

13 Platform Engineering

★ Capstone